

DYNAMIC MARKET DESIGN IN LARGE ECONOMIES

Foundations and Applications to the Design of Dynamic Incentives

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CONTEXT

Many matching markets are inherently **dynamic** – teachers in public schools [Combe et al. (2025)], police officers [Sidibé et al. (2021)], judges, even school choice [Hahm, Park, 2025]

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2. Significant **distributional imbalances** not fully addressed by monetary compensation
 - Teacher experience lower in disadvantaged schools. [Hanushek et al. (2024); Jackson (2009)]
 - Less exp. police officers assigned to high-crime neighborhoods [Sidibé et al. (2021)]

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⇒ Need for an analysis of dynamic incentives where priorities are instruments.

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Introduction of a **new dynamic matching setting** in a continuum model

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APPLICATIONS How to design priority systems to achieve distributional objectives?

RELATED LITERATURE

STATIC MATCHING IN LARGE MARKETS

Azevedo, Leshno (JPE, 2016) – Artemov et al. (JPE Micro, 2023) – Che et al. (Ecma, 2019)

SECOND WELFARE-THEOREMS / REVELATION PRINCIPLES

Ashlagi, Shi (Management Sc, 2016) – Miralles, Pycia (JET, 2021) – Shi (Management Sc, 2021)

DYNAMIC MATCHING

Akbarpour et al. (JPE, 2019) – Sönmez, Ünver (AER, 2020) – Doval (JET, 2022)

MATCHING WITH CONSTRAINTS.

Suzuki et al. (AAMAS, 2018) – Hafalir et al. (2024) – Combe et al. (2022)

DYNAMIC INCENTIVES AND DESIGN

Bergemann, Välimäki (JEL, 2019) – Guo, Hörner (2018)

EMPIRICAL MATCHING

Ederer (2026) – Galichon et al. (2022)

THE DYNAMIC FRAMEWORK

Definition and Specific Features

DYNAMIC FRAMEWORK

Overlapping generations & discrete time

ECONOMY Finite set of schools $\mathcal{S} = \{1, \dots, S\}$ - capacities $(q_s)_{s \in \mathcal{S}}$
Mass $\frac{1}{A}$ of new agents born each t who live A periods (total mass 1 of agents each t)

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$\hookrightarrow$ initial score and flow payoff for each school – drawn i.i.d from atomless $p \in \Delta(\Theta)$

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↪ **Transitions** For generation a , conditional on previous type (r, u) and assignment:

$$Q_a(\cdot \mid r, u, s) \in \Delta(\mathcal{R} \times \mathcal{U}).$$

↪ Assumption: Set of types with indifferences is of measure 0.

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$$\mu_{t,a} : \mathcal{R} \times \mathcal{U} \rightarrow \mathcal{S} \cup \{\emptyset\}.$$

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↪ (pre-)matchings and initial conditions ($\eta_{0,a}$ for agents of age $a > 1$ at $t = 0$)

↪ **distributions** over $\mathcal{R} \times \mathcal{U}$:

$\eta_{t,1}(\cdot | \mu) = p$ & $\eta_{t,a}(\cdot | \mu)$ obtained from $\eta_{t-1,a-1}$ via $Q_{a-1}(\cdot | r, u, \mu_{t-1,a-1}(r, u))$

SOLUTION CONCEPT

Stationary market-clearing cutoffs

STATIC CASE: A DETOUR

- Consider a single generation, i.e., $A = 1$.
- The economy reduces to repeated independent static environments.
- Time indices are suppressed: we omit t and a from the notation.

This corresponds to the static framework of Azevedo and Leshno (JPE, 2016).

STATIC CASE: MARKET CLEARING CUTOFFS

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- Solution = individual demand $d(P \mid r, u)$

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AGGREGATE DEMAND

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Proposition (Azevedo–Leshno)

The set of market-clearing cutoffs is a non-empty complete lattice.

Under regularity conditions on η , there is a unique market-clearing cutoff.

Key properties: Aggregate demand satisfies

- **Monotonicity:** $D_s \searrow$ in P^s
- **Gross substitutes:** $D_s \nearrow$ in P^{-s}

BACK TO THE DYNAMIC CASE

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DYNAMIC MARKET CLEARING $\mathbf{P}$ clears the market if $\forall t$, P_t is market-clearing for D_t .

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Proposition 1.

Assume the distribution is regular. The equilibria under the DA mechanism coincide with dynamic market-clearing cutoffs.

► proof

↪ All these equilibria are “truthful”

↪ Truthful equilibrium: σ = rank-ordered list induced by flow payoffs + continuation values

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- ↪ Tractable solution concept.

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(Sanity Check)

Under regularity of the distribution, the set of **stationary equilibria** under the DA mechanism coincides with the set of **stationary** market-clearing cutoffs.

REMARK Given $\{\eta_{a,0}\}_a$, an eq. generates sequences $\{\eta_{t,a}\}_t$ and $\{\mu_{t,a}\}_t$. A time-independent strategy is a **stationary equilibrium** if there exists $\{\eta_a\}_a$ such that, when $\{\eta_{a,0}\}_a = \{\eta_a\}_a$, it is an equilibrium and those sequences are constant.

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- Our solution concept characterizes stationary outcomes under repeated DA.
- What are its structural properties?

STANDARD OBJECTS?

STATIONARY DEMAND Complexities in the stationary environment

$$D^s(P) := \sum_{a=1}^A \frac{1}{A} \eta_a(\{(r, u) \mid s = d_a(P \mid r, u)\} \mid d(P))$$

Stationary demand $\rightarrow$ depends on P both through individual demand and through the measure induced by the (pre-)matching generated by P .

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BUT...

FAILURE OF GROSS SUBSTITUTES

SCHOOLS s_1, s_2

AGENTS 2 age groups, $a \in \{L, H\}$ with

$$u_{s_1} > 0 > u_{s_2}$$

Q s.t. attending s_2 at $a = L$ provides a **bonus** for s_1 at $a = H$

→ P_1 can be so high that the bonus from attending s_2 is useless → $P_1 \nearrow \Rightarrow D_2 \searrow$

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- Positive mass of agents have preference reversal
($s_2 \succ s_1$ as L and $s_1 \succ s_2$ as H)
- Two choices when L (if access to both schools):
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As P_1 increases, bet becomes riskier! More safe choice! $\rightarrow P_1 \nearrow \Rightarrow D_1 \nearrow$

► Details

FAILURE OF STRUCTURAL PROPERTIES

↪ Failure of Generic Uniqueness

Stationary market-clearing cutoffs need not be unique.

↪ Anti-Lattice Property

If $\sum_s q_s < 1$ (+ regular distrib.): Any two stationary market-clearing cutoffs are not pointwise ordered.

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 $\Rightarrow$ Intricate eq. computation $\rightarrow$ **notion of approximate eq = (ε, η) -eq**
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+ dynamic propagation of these difficulties – e.g. **large future impact of small deviations**

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Proposition.

Let σ a rich/regular Nash equilibrium of the (continuum) game induced by DA under a regular transition kernel Q ; then σ is an ex-post $(\varepsilon_N, \varepsilon_N)$ -equilibrium of the finite market game with N agents per generations, with $\varepsilon_N \xrightarrow{N \rightarrow \infty} 0$.

NORMATIVE

Implementation Principle

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- ↪ **Transitions** For generation a , conditional on previous type u and assignment:

$$Q_a^\star(\cdot \mid u, s) \in \Delta(\mathcal{U}).$$

- ↪ Assumption: Set of types with indifferences is of measure 0.

GENERAL MECHANISMS – SCHOOL ASSIGNMENTS

$$\phi: A \times M \times H \times Y \times G \rightarrow \mathcal{S} \cup \{\emptyset\}$$

- M : cheap-talk message space (compact equipped with its Borel σ -algebra.)

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REMARK We are already imposing anonymity, time-invariance and aggregativeness.

FEASIBILITY Restrict attention to feasible (+ measurable) mechanisms, i.e., for any $\zeta \in G$

$$\int \mathbb{1}_{\phi(\cdot, \zeta)=s} d\zeta \leq q_s$$

► Example

DYNAMIC GAME INDUCED BY MECHANISM ϕ

- Agents report message $m \in M$.

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⇒ **stationary equilibrium allocation** are induced by stationary equilibria.

QUESTION

Given a stationary equilibrium allocation π induced by a mechanism ϕ :

Can we implement it by appropriately designing agents' priorities and simply running DA?

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Can we implement it by appropriately designing agents' priorities and simply running DA?

Can we construct:

- distribution over scores $\hat{p} \in \Delta(\mathcal{R})$,
- transition kernels over scores $\hat{Q}(\cdot | r, s) \in \Delta(\mathcal{R})$

such that a stationary equilibrium outcome of DA induces π ?

Where measures over types $(r, u) \in \mathcal{R} \times \mathcal{U}$ is $p = \hat{p} \otimes p^*$ and $Q_a(\cdot | r, u, s) = \hat{Q}_a(\cdot | r, s) \otimes Q_a^*(\cdot | u, s)$,

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Yes, under two axioms.

AXIOMS

A1. NO DEPENDENCE ON MESSAGE RECORDS. The mechanisms does not depend on the private history of cheap-talk messages, i.e.,

$$\phi(\cdot, h) = \phi(\cdot, h') \quad \forall h \sim h',$$

where $h \sim h'$ means that h and h' differ only in their cheap-talk private history.

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~ Mechanisms used in practice

↪ A1. may be dropped if transition kernels over scores can depend on message records.

A2. NON-EXCLUSION: Given any aggregate reports $\zeta \in G$, any school s which capacity is not saturated is accessible to all, i.e.,

$$\int \phi(\cdot, \zeta) d\zeta < q_s \Rightarrow \forall (a, h, y) : \exists m \in M, \phi(a, m, h, y, \zeta) = s.$$

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↪ A2 can be removed if we allow capacity reduction.

GENERAL MECHANISM TO MARKET CLEARING CUTOFFS

Theorem.

Let ϕ be a mechanism satisfying Axioms A1 and A2, and let π be a stationary equilibrium allocation.

Then there exist a **distribution over scores** $\hat{p} \in \Delta(\mathcal{R})$ and a **transition kernel over scores** $\hat{Q}_a(\cdot \mid r, s) \in \Delta(\mathcal{R})$ under which there is a vector of **stationary market-clearing cutoff** which **induces an allocation equivalent to π** .

REMARK π and π' are **equivalent** if their induced distributions over $A \times \mathcal{U} \times S$ are the same.

GENERAL MECHANISM TO MARKET CLEARING CUTOFFS

Theorem.

Let ϕ be a mechanism satisfying Axioms A1 ~~and A2~~, and let π be a stationary equilibrium allocation.

Then there exist a capacity vector $q' \leq q$, a distribution over scores $\hat{p} \in \Delta(\mathcal{R})$ and a transition kernel over scores $\hat{Q}(\cdot | r, s) \in \Delta(\mathcal{R})$ under which there is a vector of stationary market-clearing cutoff which induces an allocation equivalent to π .

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GENERAL MECHANISM TO MARKET CLEARING CUTOFFS

Corollary 1.

Let ϕ a mechanism satisfying Axioms A1 and A2. Let π be a stationary equilibrium allocation.

Assume distributions are regular. There exist a **distribution over scores** $\hat{p} \in \Delta(\mathcal{R})$ and a **transition kernel over scores** for which there is an equilibrium under DA yielding an allocation equivalent to π .

► Intuition

► Proof Sketch

APPLICATIONS

Tradeoffs in the Design of Dynamic Incentives

DYNAMIC INCENTIVES FOR DISTRIBUTIONAL OBJECTIVES – OVERVIEW

CONTEXT Can dynamic incentives ensure a more equitable distribution of experience?

Several applications using our implementation principle:

DYNAMIC INCENTIVES FOR DISTRIBUTIONAL OBJECTIVES – OVERVIEW

CONTEXT Can dynamic incentives ensure a more equitable distribution of experience?

Several applications using our implementation principle:

- ↪ Structural tradeoffs between using dynamic incentives & other desirable properties
 - E.g. "seniority bonuses"/"squatting rights" ($\Rightarrow$ older teachers assigned desirable schools)
 - in a relevant environment, cannot implement non-assortative matchings
 $\approx$ empirical regularity
- ↪ Given **distributional constraints**, show we can design priorities to achieve **constrained-efficiency under DA**.
 - Proof suggests systematic way to design dynamic incentives from objectives (in progress)
 - Idea:
 - (i) Build a pseudo-market equilibrium pricing the constraints (inspired by Echenique et al. 2021)
 - (ii) Apply implementation principle to get a simple mechanism where we price schools.

CONCLUDING REMARKS

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Still preliminary research project: What's missing?

1. Our “implementation principle” is a first step toward optimal design.

- ↪ So far: way to identify priority systems yielding constrained efficiency
- ↪ What do they look like with real-world distributional objectives?

CONCLUDING REMARKS

Still preliminary research project: What's missing?

1. Our “implementation principle” is a first step toward optimal design.

- ↪ So far: way to identify priority systems yielding constrained efficiency
- ↪ What do they look like with real-world distributional objectives?

2. Our framework can be used to do demand estimation.

- ↪ Artemov, Che and He 2023, Fack, Grenet and He 2019, Dur et al. 2022
- ↪ Work in progress...

RELATED LITERATURE

STATIC MATCHING IN LARGE MARKETS

Azevedo, Leshno (JPE, 2016) – Artemov et al. (JPE Micro, 2023) – Che et al. (Ecma, 2019)

SECOND WELFARE-THEOREMS / REVELATION PRINCIPLES

Ashlagi, Shi (Management Sc, 2016) – Miralles, Pycia (JET, 2021) – Shi (Management Sc, 2021)

DYNAMIC MATCHING

Akbarpour et al. (JPE, 2019) – Sönmez, Ünver (AER, 2020) – Doval (JET, 2022)

MATCHING WITH CONSTRAINTS.

Suzuki et al. (AAMAS, 2018) – Hafalir et al. (2024) – Combe et al. (2022)

DYNAMIC INCENTIVES AND DESIGN

Bergemann, Välimäki (JEL, 2019) – Guo, Hörner (2018)

EMPIRICAL MATCHING

Ederer (2026) – Galichon et al. (2022)

Thank You!

Proof of Proposition 1 – 1/2

Fix $\mathbf{P} = \{P_t\}_t$ dynamic market-clearing cutoffs.

Under DA, build an equilibrium which induces sequence $\mathbf{P}$.

- $\sigma_{a,t}(r, u)$ ranks first the most preferred school (flow payoff + cont. value) within accessible schools given sequence $\mathbf{P}$.
- Given σ , recursively compute aggregate demand at each time t . By defn, P_t is a market-clearing cutoff at time t and the unique by regularity (hence, the smallest).

$\Rightarrow$ Under DA, the sequence induced is $\mathbf{P}$ and σ is an equilibrium.

Proof of Proposition 1 – 2/2

Fix an equilibrium σ under DA and $\mathbf{P}$ the induced sequence of market-clearing cutoffs.

Show: $\mathbf{P}$ are dynamic market-clearing cutoffs.

- By optimality of σ , at each t , (r, u) obtains his *true* most favorite school (flow payoff + cont. value) within those that are accessible under P_t .
- Market-clearing holds at each time t under true demands given sequence $\mathbf{P}$.

$\Rightarrow$ $\mathbf{P}$ are dynamic market-clearing cutoffs.

FAILURE OF MONOTONICITY

SCHOOLS s_1, s_2 – capacities q_1, q_2 with $q = q_1 + q_2 < 1$

AGENTS 2 age groups, $a \in \{L, H\}$ with :

- Scores $\sim U[0, 1]$ at each age + squatting rights
- Utility $u_a^s(t) = \bar{u}_a^s(t) + \varepsilon_a^s$,
 - $t \in \mathcal{T}$, finite set of types
 - $\bar{u}_a^s(t)$ type-specific value of s at age a
 - ε_a^s individual-specific random shock – $\mathbb{E}(\varepsilon_a^s) = 0$, for all a, s

NB Shock realizations are revealed to the agent at birth

ASSUMPTION There exists a type $t^\star \in \mathcal{T}$ with positive mass such that

$$r = \frac{\bar{u}_L^{s_1}(t^\star) - \bar{u}_L^{s_2}(t^\star)}{\bar{u}_H^{s_2}(t^\star) - \bar{u}_H^{s_1}(t^\star)} \in [0, 1]$$

↪ Reversal in type-specific value of schools

FAILURE OF MONOTONICITY

Assume the variance of the idiosyncratic shocks vanish. There exists an open set of capacity vectors such that there are **multiple stationary market-clearing cutoffs** at these capacities.

INSURANCE t^* -agents with access to both schools at L

($s_2 > s_1$ as L and $s_1 > s_2$ as H)

- if $u_L^{s_2}(t^*) + P_1 u_H^{s_2}(t^*) + (1 - P_1) u_H^{s_1}(t^*) < u_L^{s_1}(t^*) + u_H^{s_1}(t^*)$
 $\Rightarrow$ **Insurance behavior:** demand s_1 at L
- When shock vanishes
 - $\hookrightarrow$ insurance for $P_1 > r$; no insurance for $P_1 < r$

FAILURE OF MONOTONICITY

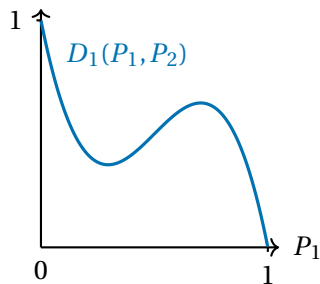
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AGGREGATE DEMAND When noise is low, a small increase in P_1 raises the insurance behavior of t^* -agents
 $\Rightarrow$ demand for $s_1 \uparrow \Rightarrow$ **Non-monotonic aggregate demand**



Equilibrium definition

MARKOV DECISION PROBLEM Given strategy σ and the induced distribution (ζ_t) , each agent solves a **Markov decision problem**:

$$U_t(u, a, y, h) := \max_{m \in M} u(s) + \int_{\mathcal{U}} \int_Y U_{t+1}(\tilde{u}, a, \tilde{y}, h \uplus (m, y, s)) dQ(\tilde{u}|u, s) d\tilde{y}$$

where $s = \varphi_t(m, a, y, h | \zeta_t)$.

(with convention $U_t(\cdot, A+1, \cdot, \cdot) \equiv 0$).

Strategy σ is an **equilibrium** if the prescribed messages solve the MDP, i.e.

$$\sigma_t(u, a, y, h) \in \arg\max_{m \in M} u(s) + \int_{\mathcal{U}} \int_Y U_{t+1}(\tilde{u}, a, \tilde{y}, h \uplus (m, y, s)) dQ(\tilde{u}|u, s) d\tilde{y}$$

where $s = \varphi_t(m, a, y, h | \zeta_t)$.

THE FINITE ECONOMY

FINITE ECONOMY Finite set of schools $\mathcal{S} = \{1, \dots, S\}$ – **finite capacities** $(q^{s,n})_{s \in \mathcal{S}}$
New generation with **n agents** born at each t – **types drawn i.i.d. from p**
For generation $a > 1$, conditional on previous types and assignments
– **new types drawn i.i.d. from Q_{a-1}**

DISTRIBUTIONS Initial conditions $(\eta_{0,a}^n)_{a=1 \dots A}$ for agents of age $a > 1$ at $t = 0$

→ **Convergence** of a finite economy to a continuum economy :

$$\frac{1}{nA} q^{s,n} \xrightarrow{n \rightarrow \infty} q^s \quad \& \quad \eta_0^n \xrightarrow{n \rightarrow \infty} \eta_0$$

DA MECHANISM IN THE FINITE WORLD

- Agents report rank-ordered lists $\ell \in \mathcal{L}$.
- Strategies are maps from **histories**, time, generation and types to lists $\ell \in \mathcal{L}$.
 - ↪ **Histories = past assignments, past scores**
- Aggregate demand $D_t(\cdot)$ is **recursively** computed from reported lists.
 - ↪ **Random variable now**
- The mechanism **recursively** selects the smallest market-clearing cutoff P_t^-
 - ↪ **Random variable now**
 - ↪ **One can take an arbitrary selector**

A strategy induces a random sequence of market-clearing cutoffs $\mathbf{P}$.

APPROXIMATE EQUILIBRIUM

FINITE ECONOMY Difficulties: inference & market power

- Need to specify beliefs over types and assignments of others over their lifetime as function of current type and history
 - Current report impacts tomorrow types and matchings $\rightarrow$ externalities
- $\Rightarrow$ Intricate eq. computation $\rightarrow$ notion of approximate eq = (ε, η) -eq

Definition. A strategy profile is an **ex-post (ε, η) -equilibrium** if the probability that some type (r, u) realized at some date t with age a has an ε -profitable deviation is at most η .

$\hookrightarrow$ Interest of the continuum $\rightarrow (\varepsilon, \eta)$ -equilibrium is a good approximation

FINITE TO CONTINUUM – CONVERGENCE

Proposition.

Let σ a rich/regular Nash equilibrium of the (continuum) game induced by DA. Let the transition kernel Q be regular; then σ is an ex-post $(\varepsilon_N, \varepsilon_N)$ -equilibrium of the finite market game with N agents per generations, with $\varepsilon_N \xrightarrow{N \rightarrow \infty} 0$.

- Rich: the distribution on scores & ROLs, ν , (induced by σ and η) has **full support**
- Regular equilibrium: the distribution on scores & ROLs, ν , is s.t. conditional on ℓ , the measure of scores at which demand is **discontinuous** vanishes uniformly in P
- Regular transitions: contractivity of the Markov transition kernel for the 1-Wasserstein distance

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STEPS OF THE PROOF

- "Static" (period by period) almost sure convergence of market outcomes (demands and cutoffs) and vanishing market power – richness/regularity
- Uniform convergence over time – regularity on transitions
- Vanishing gains from deviation

EXAMPLE

REPEATED DA Allocation rule: $\pi^{DA} : \mathcal{L} \times R \times \Delta(\mathcal{L} \times R) \rightarrow S \cup \{\emptyset\}$

↪ Messages: $M = \mathcal{L}$ (rank-ordered lists)

↪ Priorities: in $R := [0, 1]^S$ drawn sequentially from $\hat{p}, \hat{Q}(\cdot | r, s) \in \Delta(R)$

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$$\begin{array}{ll} r_1 : Y \rightarrow R & \text{s.t. } \mathbb{P}_{y_1 \sim \mathcal{U}[0,1]}(r_1(y_1) \in A) = \hat{p}(A) \\ r_2 : (S \times Y) \times Y \rightarrow R & \text{s.t. } \mathbb{P}_{y_2 \sim \mathcal{U}[0,1]}(r_2((s_1, y_1), y_2) \in A) = \hat{Q}(A \mid r_1(y_1), s_1) \\ \vdots & \\ r_a : (S \times Y)^{a-1} \times Y \rightarrow R & \text{s.t. } \mathbb{P}_{y_a \sim \mathcal{U}[0,1]}(r_a(h, y_a) \in A) = \hat{Q}(A \mid r_{a-1}(h_{a-1}), s_{a-1}) \end{array}$$

AGGREGATES $\xi : \Delta(A \times \mathcal{L} \times H \times Y) \rightarrow \Delta(\mathcal{L} \times R) \rightsquigarrow \xi_a(g) := (\text{Id}, r_a) \# g_a; \xi = \sum_a \frac{1}{A} \xi_A$

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Equivalent Mechanism: $\phi^{DA}(a, \ell, h, y, g) := \pi^{DA}(\ell, r_a(h, y); \xi(g))$

INTUITION FOR IMPLEMENTATION PRINCIPLE

Basic idea:

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 $\hookrightarrow$ Scores $\geq 1/2$ for school s if and only if s is in the menu of $(1, \emptyset, y)$
3. Distribution $\hat{p}$ = distribution of $f(y)$.
4. Difficult = ensure agents get same menus **and same distribution over future menus!**
 - Idea: Encode histories and current shocks into the score.

THEOREM – SKETCH OF THE PROOF

MENU REPRESENTATION [Hammond, 1979], [Ashlagi, Shi, 2016], [Shi, 2021]

Given aggregate reports ζ , as if agent (a, h, y) picks most preferred school in a menu:

$$\mathcal{M}(a, h, y) := \bigcup_{m \in M} \{\phi(m, a, h, y \mid \zeta)\}$$

↪ Time-independent menus because stationary state.

↪ Uses no message recording (A1)

THEOREM – SKETCH OF THE PROOF

MENU REPRESENTATION [Hammond, 1979], [Ashlagi, Shi, 2016], [Shi, 2021]

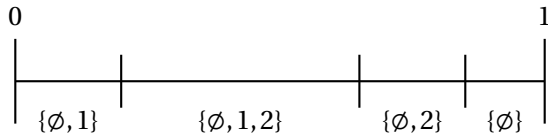
Given aggregate reports ζ , as if agent (a, h, y) picks most preferred school in a menu:

$$\mathcal{M}(a, h, y) := \bigcup_{m \in M} \{\phi(m, a, h, y \mid \zeta)\}$$

↪ Time-independent menus because stationary state.

↪ Uses no message recording (A1)

↪ E.g., For $a = 1$, menus are a function of $y \in [0, 1]$ (recall $h = \emptyset$) and so described by



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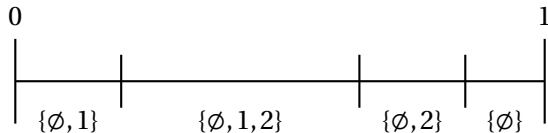
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⇒ We use the representation in terms of menus.

THEOREM – SKETCH OF THE PROOF

IMPLEMENTATION VIA STATIONARY MARKET-CLEARING Focus 2 schools.

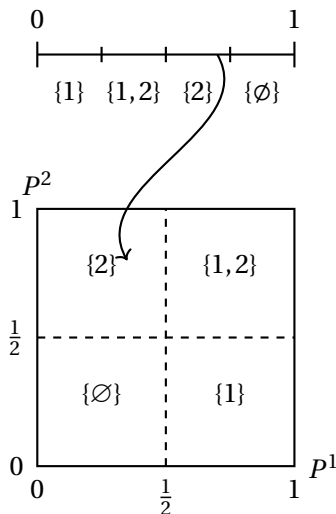
- Cutoffs: $P^s = 1/2$ for all s

Consider first $A = 1$.

Goal: Build a map $f: Y \rightarrow [0, 1]^S$ so that $f(y)$ behaves exactly as $(\emptyset, y)$.

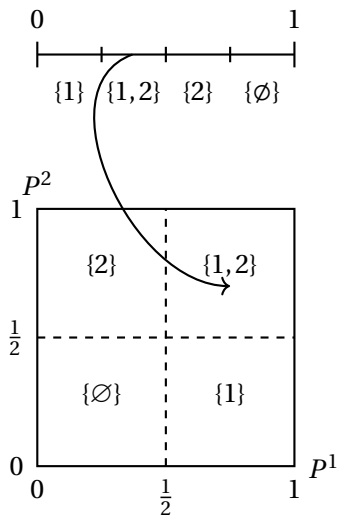
THEOREM – SKETCH OF THE PROOF

GRAPHICAL REPRESENTATION OF f



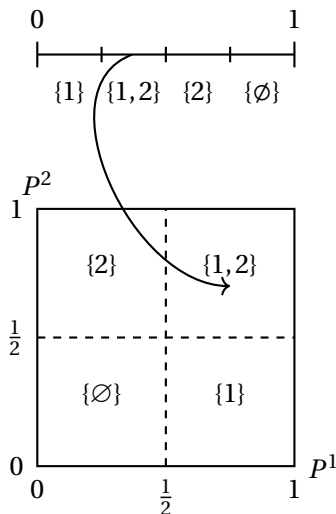
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$\Rightarrow$ This defines a map $f : Y \rightarrow [0, 1]^S$

THEOREM – SKETCH OF THE PROOF

CONCLUSION FOR $A = 1$

Given map $f : Y \rightarrow [0, 1]^S$ defined by

$$\frac{1}{2} \mathbb{1}_{\{s \in \mathcal{M}(1, \emptyset, y)\}},$$

we have

$$s \in \mathcal{M}(1, \emptyset, y) \iff r^s := f^s(y) \geq 1/2 = P^s.$$

$\implies f(y)$ and y have the same menus and so behave in the same way.

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Define the distribution over r as the distribution over $f(y)$.

$\hookrightarrow$ I.e., the pushforward of the distribution of y by f

$\implies$ The allocation is the same under cutoffs $(1/2, 1/2)$ and under mechanism ϕ .

THEOREM – SKETCH OF THE PROOF

Consider $A = 2$.

Goal: Build $f : H \times Y \rightarrow [0, 1]^S$ s.t. agents with type (h, y) behave in the same way as $f(h, y)$.

Issue: If we use the same mapping f :

Menus as $(\emptyset, y)$ and as $f(\emptyset, y)$ are the same **but not distribution over future menus!**

THEOREM – SKETCH OF THE PROOF

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Idea: Encode histories and current shock (h, y) in the score:

- Bijection (Borel isomorphism) : $\xi : (h, y) \rightarrow [0, 1/2]$
- Map $f : H \times Y \rightarrow [0, 1]^S$ defined by

$$\frac{1}{2} \mathbb{1}_{\{s \in \mathcal{M}(a, h, y)\}} + \xi(h, y),$$

$\hookrightarrow$ As before, $(\emptyset, y)$ and $f(\emptyset, y)$ induce identical menus.

Moreover: Given $r = f(h, y)$, we can recover (h, y) and build kernel $\hat{Q}$ to ensure that the distributions over future menus coincide for (h, y) and $f(h, y)$.

$\implies f(\emptyset, y)$ and $(\emptyset, y)$ behave in same way.

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Consider $A = 2$ (continued) What about distribution over future menus?

Need to build transitions $\hat{Q}$ appropriately.

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Need to build transitions $\hat{Q}$ appropriately. For score r and assignment s at $a = 1$

- Back out shock y from score r (injection f)

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If $\hat{p}'$ over r is the pushforward distrib. of y by f : same allocation under cutoffs $(1/2, 1/2)$ and mechanism ϕ .

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ENVIRONMENT Generation-invariant payoffs,

→ Payoffs drawn according to p^* upon arriving and remain the same afterwards.

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CONSTRAINED ALLOCATIONS Focus on allocations π in $\mathcal{C} \neq \emptyset$ satisfying a finite set of linear constraints

$$\sum_a \sum_s \int_{(u, \tilde{h}, y)} \alpha_{a,s}^c \mathbb{1}_{\{\pi(a, u, \tilde{h}, y) = s_1\}} d\tilde{\eta} \leq \beta_c$$

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→ e.g. Capacity constraints: $\int_{(u, \tilde{h}, y)} \sum_a \sum_s \mathbb{1}\{\pi(a, u, \tilde{h}, y) = s\} d\tilde{\eta} \leq q^s$

$\tilde{h}$: history of draws; $\tilde{\eta}$: distribution over primitive characteristics $(u, \tilde{h}, y)$

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↪ e.g. **Distributional** constraints: **Bound on the mass of A -agents at s_1** $\int_{(u, \tilde{h}, y)} \mathbb{1}\{\pi(A, u, \tilde{h}, y) = s_1\} d\tilde{\eta} \leq \beta$

EFFICIENT INEQUALITY-REDUCING MECHANISMS

Proposition.

There exist a **distribution over scores**, **transition kernels over scores** and **capacities $q' \leq q$** under which there is a vector of **stationary market-clearing cutoffs** that implements a **constrained efficient allocation**.

↪ **Constrained efficiency**: $\nexists \pi'$ in $\mathcal{C}$ which Pareto-dominates π [▶ Def.](#)

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Corollary.

Assume regular distributions. There exist a **distribution over scores** and **transition kernels over scores** s.t. under DA, some stationary equilibrium allocation is constrained-efficient.

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- Given distributional constraints, proof suggests a systematic way to construct $\hat{p}$ and $\hat{Q}$.

[► Back](#)

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Idea:

- (i) Build a pseudo-market equilibrium where we price the constraints

Inspired by Echenique et al. 2021

- (ii) Apply implementation principle to get a simple mechanism where we price schools.

EFFICIENT INEQUALITY-REDUCING MECHANISMS – PROOF IDEA

We assume that each agent draws i.i.d. a budget $B \sim U[0, 1]$.

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↪ Based on the contribution of each generation to constraints.

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2. When agents pick most preferred trajectory $(s_1, \dots, s_A)$ in their budget set:

$$B \geq P_{1,s_1}^\star + \dots + P_{A,s_A}^\star$$

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3. $P_c > 0$ implies that constraint c is binding.

REMARK Idea of pricing constraints is inspired by Echenique et al., 2021.

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Constrained efficient allocation obtained through pseudo-market equilibrium.

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There exists a Pseudo-market constrained-equilibrium and it is constrained efficient.

- ↪ This mechanism has a stationary equilibrium allocation which is constrained efficient.
- ↪ By our **implementation principle**: Eq. allocation implementable through DA with appropriate design of scores.

DYNAMIC INCENTIVES FOR DISTRIBUTIONAL OBJECTIVES

PREFERENCES Drawn from $\mathcal{U} = \{(u, \delta) \in [0, 1]^S \times [0, 1] : u_1 > u_2 > \cdots u_S \geq 0 \quad \& \quad \sum_s \delta^{S-s} u_s \geq 0\}$
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BALANCED ENVIRONMENT

- Number of generations = Number of schools, i.e., $|A| = |S|$
 - Uniform capacities $q_s = \frac{1}{S}, \forall s$
- Well-defined assortative matching

$$\begin{pmatrix} 1 & 2 & \dots & A \\ S & S-1 & \dots & 1 \end{pmatrix}$$

FRAGILITY OF NON-ASSORTATIVE MATCHINGS – EXAMPLE

EXAMPLE $|S| = |A| = 3$, $u_3 = -1$, $u_2 = 0.5$, $u_1 = 3$, discount factor $\delta \sim \mathcal{U}([x, 1]) \rightarrow$ Can we implement matching

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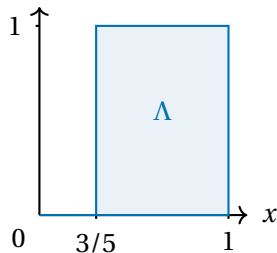
$$\begin{pmatrix} 1 & 2 & 3 \\ s_2 & s_3 & s_1 \end{pmatrix}$$

- $\hookrightarrow$ for $x \geq 3/5$ every agent is **patient** : prefers sequence of matches $s_2 s_3 s_1$ to $s_2 s_2 s_2$
- $\hookrightarrow$ Initial score $\sim \mathcal{U}(0, 1)$
- $\hookrightarrow$ Score bonuses:
 - (i) per year of experience
 - (ii) for generation-2 agents enrolled in school s_3
 - (iii) Squatting rights.

FRAGILITY OF NON-ASSORTATIVE MATCHINGS – EXAMPLE

STATIONARY MARKET-CLEARING:

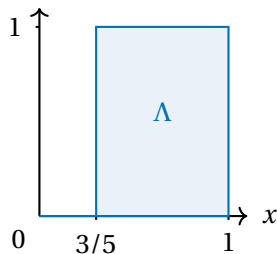
$$\Pi^*(x) = \left\{ (1 - \lambda) \begin{pmatrix} 1 & 2 & 3 \\ s_3 & s_2 & s_1 \end{pmatrix} + \lambda \begin{pmatrix} 1 & 2 & 3 \\ s_2 & s_3 & s_1 \end{pmatrix}, \lambda \in \Lambda(x) \right\}$$



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We show: **Tradeoff** between using dynamic incentives & other desirable properties.

► Back

SENIORITY AND ASSORTATIVENESS

Definition. A mechanism satisfies **seniority** if, for all scores realisations, whenever an agent of age a has access to a school s , all agents of age $a' > a$ also have access to s .

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↪ Relevant in practice

↪ The mechanism in the previous example satisfies seniority.

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Let ϕ a mechanism satisfying axioms A1 and A2. Assume ϕ satisfies **seniority** and the distribution over agents type (u, δ) has **full support**. Let π be a stationary equilibrium allocation.

Then, π is **assortative**.

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Then, π is **assortative**.

Idea:

- (i) By implementation principle we can focus on stationary market-clearing cutoffs;
- (ii) Regardless of the non-assortative matching you consider, impatient agents remain unsatisfied, generating a mass leakage that is inconsistent with market clearing.

PROPOSITION 3 – SKETCH OF THE PROOF

IMPLEMENTATION PRINCIPLE → Can focus on score mechanism

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BUILDING THE ALLOCATION Recursive construction

→ intuition with $|A| = |S| = 3$ + deterministic score

⇒ agents are choosing trajectories at age $a = 1$

Step 1. Age-3-agents assigned to s_1

- Otherwise: young agents in s_1 $\xRightarrow{\text{seniority}}$ s_1 accessible to all age-3-agents
⇒ some agents were taking trajectory $xy s'$ with $s' \neq s_1 \Rightarrow$ profitable deviation

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Otherwise

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 - ↪ Impatient agents never assigned $x_{s_3 s_1}$ & can't be assigned to $x_{s_2 s_2}$ (step 1)

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- Positive mass of age-1-agent in s_2
 - $\xRightarrow{\text{seniority}}$ all age-2 and older have access to $s_2 \Rightarrow x_{s_2 s_2} \in \mathcal{M}$
 - $\xRightarrow{\text{full support}}$ $\exists$ impatient agents: $x_{s_2 s_2} \succ x_{s_3 s_1}$

↪ Impatient agents never assigned $x_{s_3 s_1}$ & can't be assigned to $x_{s_2 s_2}$ (step 1)

⇒ $\exists$ trajectory in $\mathcal{M}$ that dominates $x_{s_2 s_2} \Rightarrow x_{s_2 s_1} \in \mathcal{M} \Rightarrow$ No agent picks $x_{s_3 s_1}$ in $\mathcal{M}$

PROPOSITION 3 – SKETCH OF THE PROOF

IMPLEMENTATION PRINCIPLE → Can focus on score mechanism

BUILDING THE ALLOCATION Recursive construction

→ intuition with $|A| = |S| = 3$ + deterministic score

⇒ agents are choosing trajectories at age $a = 1$

Step 2. Age-2-agents assigned to s_2

Otherwise

- Positive mass of age-2-agents in $s_3 \xrightarrow{\text{step 1}} \text{take } x_{s_3 s_1}$
 - $x_{s_3 s_1} \in \mathcal{M}$, menu of these agents

- Positive mass of age-1-agent in s_2

seniority
⇒ all age-2 and older have access to $s_2 \Rightarrow x_{s_2 s_2} \in \mathcal{M}$

full support
⇒ $\exists$ impatient agents: $x_{s_2 s_2} \succ x_{s_3 s_1}$

↪ Impatient agents never assigned $x_{s_3 s_1}$ & can't be assigned to $x_{s_2 s_2}$ (step 1)

⇒ $\exists$ trajectory in $\mathcal{M}$ that dominates $x_{s_2 s_2} \Rightarrow x_{s_2 s_1} \in \mathcal{M} \Rightarrow \text{No agent picks } x_{s_3 s_1} \text{ in } \mathcal{M}$

⇒ Age-3-agents must be assigned s_3

CONSTRAINTS & EFFICIENCY

CONSTRAINED EFFICIENT ALLOCATION $\pi \in \mathcal{C}$ and $\nexists \pi' \in \mathcal{C}$ s.t. $\forall a$

$$\forall (u, \tilde{h}, y) : \sum_{a'=a}^A \left(\sum_s \mathbb{1}_{\{\pi'(a', u, \tilde{h}, y) = s\}} \delta^{a'-a} u(s) \right) \geq \sum_{a'=a}^A \left(\sum_s \mathbb{1}_{\{\pi(a', u, \tilde{h}, y) = s\}} \delta^{a'-a} u(s) \right)$$

and, for some generation, the inequality holds strictly for a positive measure of types

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